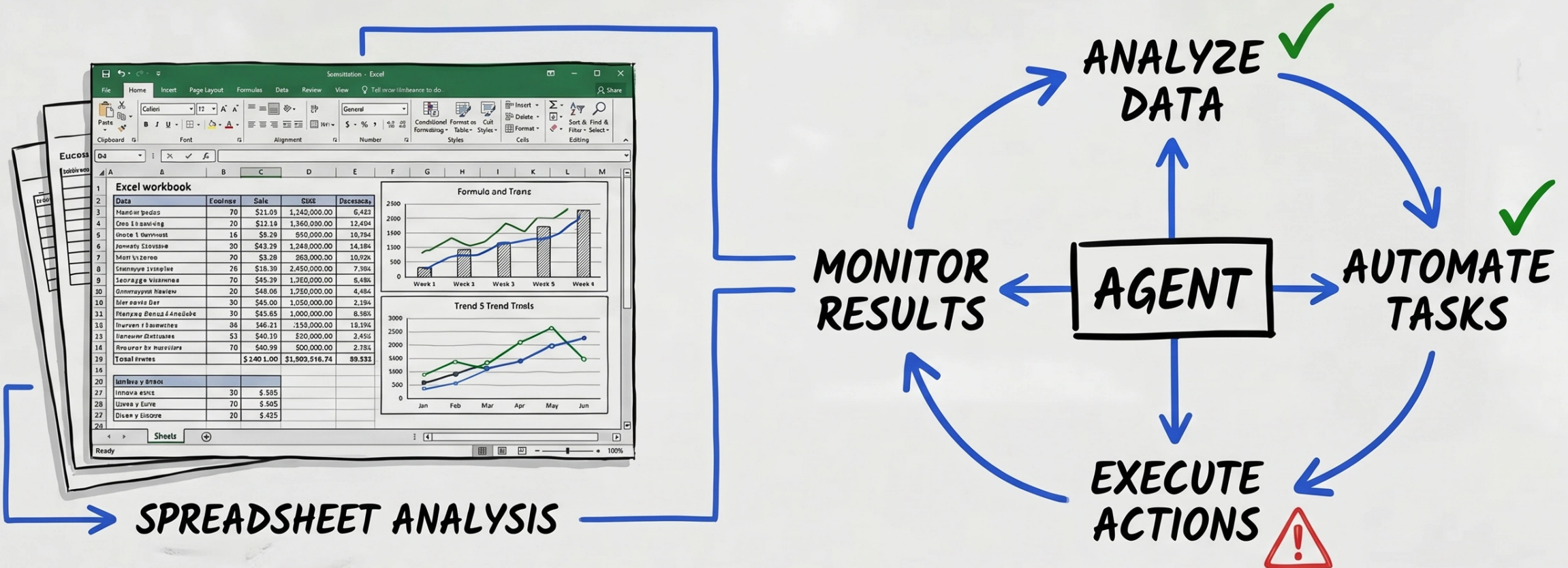


Advanced Copilot in Excel

From spreadsheet analysis to automated agent workflows

Chapter 7 — Hackathon-ready workflow building



The Spreadsheet Is the Starting Line

Branch Metrics.xlsx

A	B	C	D	E
	Deposits	Tickets	Satisfaction	Exceptions
Mr	35000	70	60%	—
Mr	10500	48	90%	0
Mr	35000	54	78%	0
Mr	42000	30	58%	0
Mr	35000	35	70%	0
Mr	10000	20	70%	~
Mr	40000	20	45%	0
Mr	30000	50	70%	0
Mr	30000	15	45%	—
Mr	7000	2	40%	—

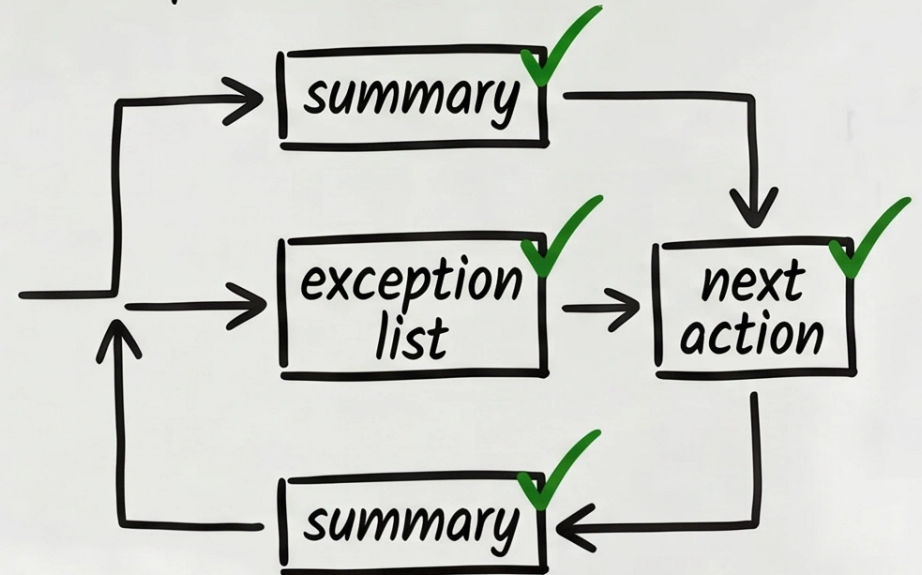
⚠ Human review stays in charge

ANALYZE

VERIFY

STANDARDIZE

Repeatable Decision Flow



Raw metrics are only the input

Copilot helps reveal patterns

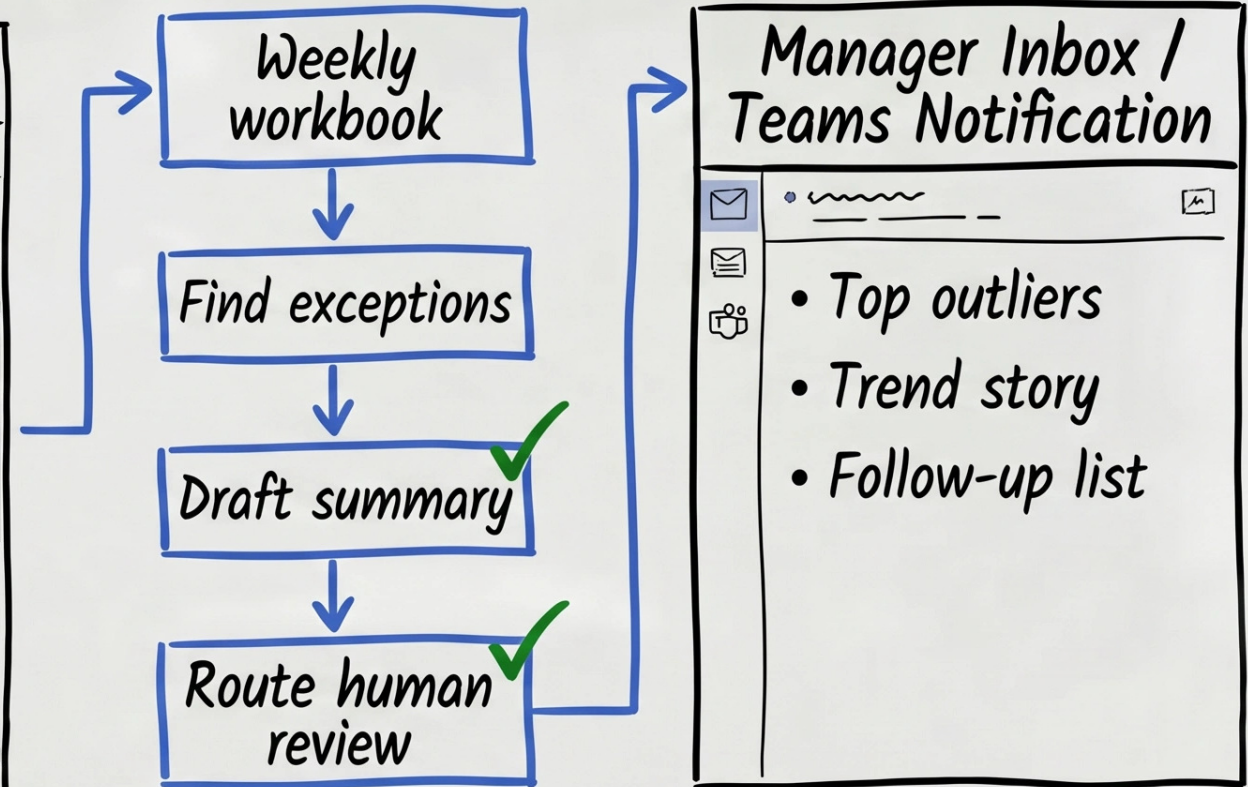
Automation turns work into a reusable workflow

Use Case: Weekly Branch Exception Review

One workbook. One weekly rhythm. One repeatable decision workflow.

BranchMetrics table

	A	B	C	D	E	F	G
		Deposits	Referrals	Tickets	Satisfaction	Digital %	Exceptions
1							
2	W	3	2	2	44	3%	2
3	W	W	—	—	W	0%	W
4	W	W	1	1	W	5%	W
5	W	W	—	—	W	1%	W
6	}	}	}	}	}	}	}



Make the Data Copilot-Ready

BranchMetrics

Merged cells



Date	Region	Branch	Sales	Profit
10/25/23	North	Branch	\$4,000	5,500
10/25/23	East	Branch	\$2,800	\$1,000
10/25/23	North	Branch	\$3,800	\$2,300
10/25/23	South	Apple	\$3,600	\$1,600
10/25/23	East	Branch	\$3,500	8,900
10/25/23	East	Aprica	\$3,000	3,450



BranchMetrics



Date	Region	Branch	Sales	Profit
10/28/23	North	Branch	\$10,500	6,200
10/25/23	East	Branch	\$30,000	\$3,000
10/25/23	South	Branch	\$18,400	\$1,500
10/26/23	North	Germany	\$14,300	\$3,800
10/27/23	East	Apple	\$20,800	\$2,000
10/28/23	South	Branch	\$10,000	9,800

Key points shown as a
handwritten checklist on the right

- ✓ One header row
- ✓ Clear column names
- ✓ Consistent dates
- ✓ Consistent categories
- ✓ No merged cells 
- ✓ Defined Excel table

Try This: Ask the Workbook

Audience members use Copilot in Excel to move from staring at rows to asking sharper business questions.

BranchMetrics					
	Branch	Region	Sales	Expenses	Profit
		Nort	10,000	13,400	3,500
		East	9,600	8,600	4,200
		East	6,200	7,200	3,300
		Nort	3,000	6,800	4,500
		East	7,000	9,300	2,300

1. Summarize trends

✓ Use column names

2. Compare markets

3. Find outliers

4. What changed?

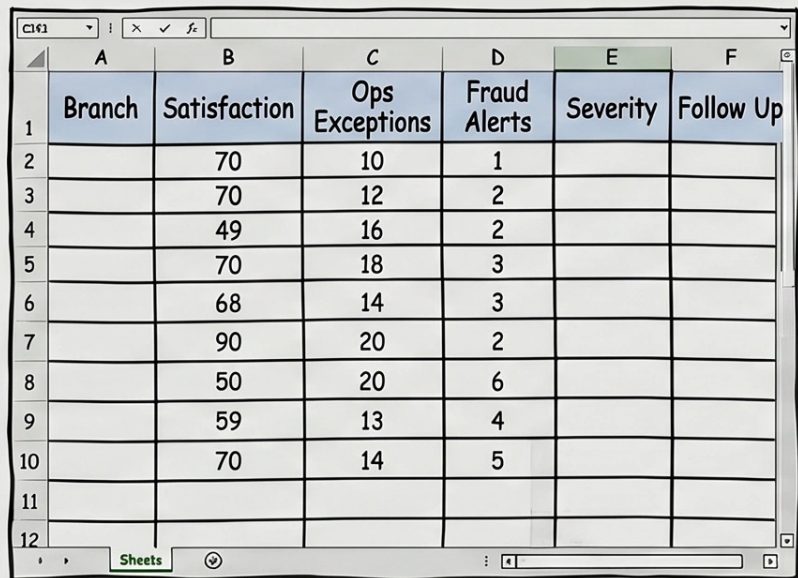
⚠ Don't trust — verify

Ask → Verify → Refine

2 minutes: choose one prompt, run it, then verify one answer against the table.

Add Judgment With Formula Columns

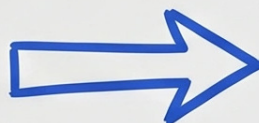
Copilot can suggest the formula — you own the judgment



A screenshot of an Excel spreadsheet with columns A through F. Column A is 'Branch', B is 'Satisfaction', C is 'Ops Exceptions', D is 'Fraud Alerts', E is 'Severity', and F is 'Follow Up'. Rows 2 through 10 contain data. Row 11 is empty. Row 12 is also empty. The spreadsheet is titled 'Sheets' at the bottom.

	A	B	C	D	E	F
1	Branch	Satisfaction	Ops Exceptions	Fraud Alerts	Severity	Follow Up
2		70	10	1		
3		70	12	2		
4		49	16	2		
5		70	18	3		
6		68	14	3		
7		90	20	2		
8		50	20	6		
9		59	13	4		
10		70	14	5		
11						
12						

raw metrics



decision-
ready
columns

SEVERITY FORMULA

=IF(FraudAlerts>2,'High',
IF(OpsExceptions>5,'Medium','Low'))

Severity = fraud + exceptions
+ customer impact



=TREND CHECK

(this week vs. prior weeks)

=FOLLOW UP?

(clear Yes / No signal)

=IF(Severity='High','Yes','No')



routine



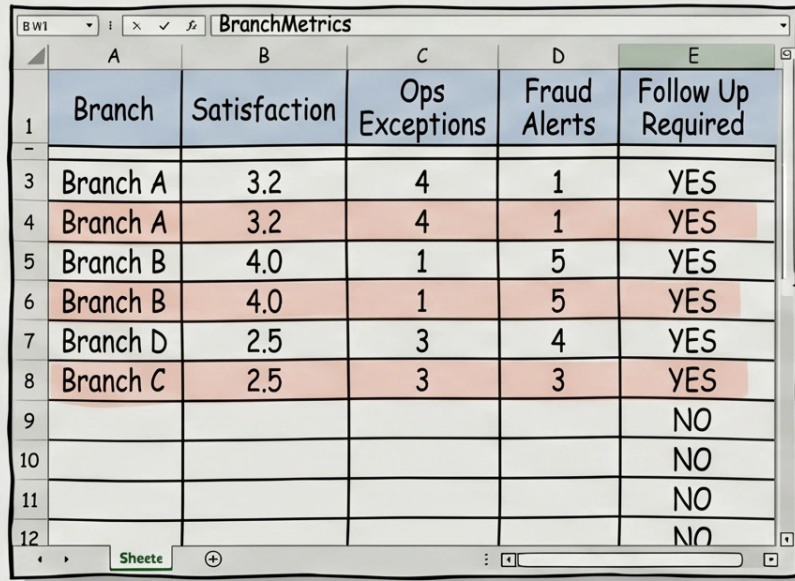
review



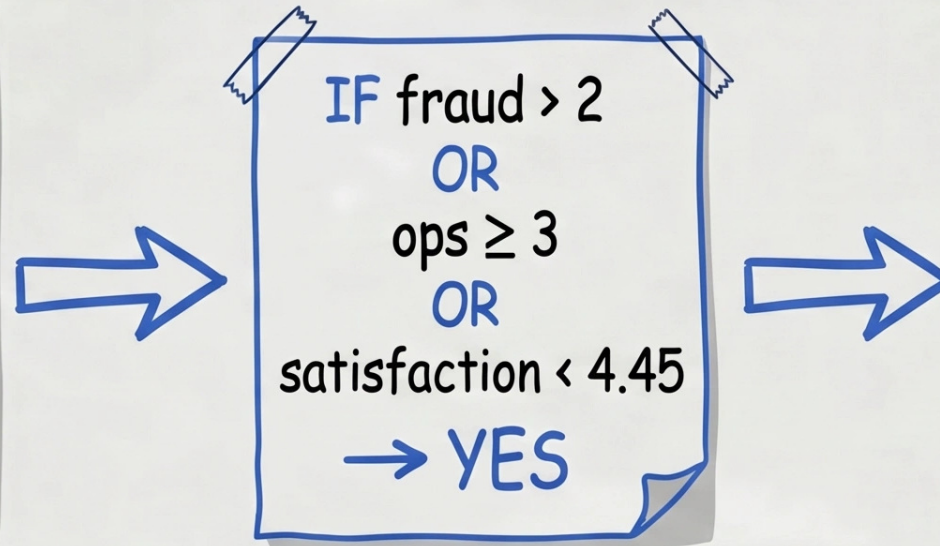
escalate

Try This: Build an Exception View

3 minutes: create or validate a Follow Up Required flag, then filter to branches needing attention.



	A	B	C	D	E
	Branch	Satisfaction	Ops Exceptions	Fraud Alerts	Follow Up Required
1					
3	Branch A	3.2	4	1	YES
4	Branch A	3.2	4	1	YES
5	Branch B	4.0	1	5	YES
6	Branch B	4.0	1	5	YES
7	Branch D	2.5	3	4	YES
8	Branch C	2.5	3	3	YES
9					NO
10					NO
11					NO
12					NO



Follow-up list

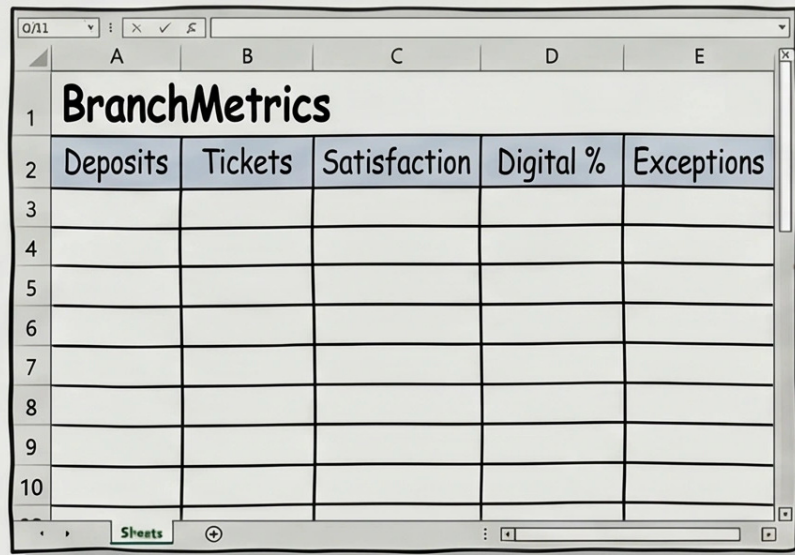
- Top risk branch
- Why flagged?
- Next owner

- ✓ validate formula
- ✓ filter results
- ⚠ verify against raw data

Run it → check it → explain it.

Advanced Analysis: Forecast, Cluster, Explain

when available, Python in Excel can help move beyond basic summaries into deeper analytical patterns, but the business user must still verify the logic and decide what action is appropriate.

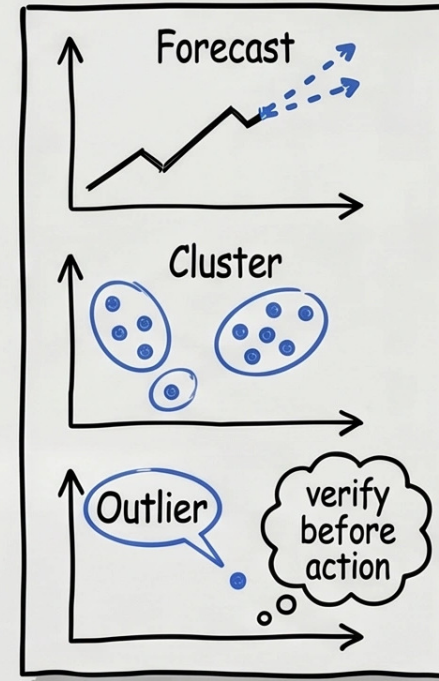


	A	B	C	D	E
1	BranchMetrics				
2	Deposits	Tickets	Satisfaction	Digital %	Exceptions
3					
4					
5					
6					
7					
8					
9					
10					

FORECAST: what may happen next?

CLUSTER: which branches behave alike?

EXPLAIN: why is this an outlier?



✓ pattern found

✓ business question answered

⚠ availability + governance vary

Advanced $\neq$ autonomous — human review still owns the decision.

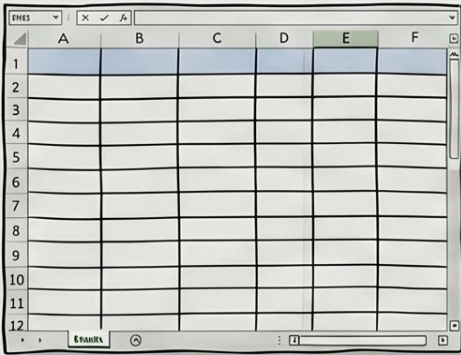
The Repeatability Gap

If the work matters, it cannot live only as a one-time chat.

One-Time Analysis



great answer
once

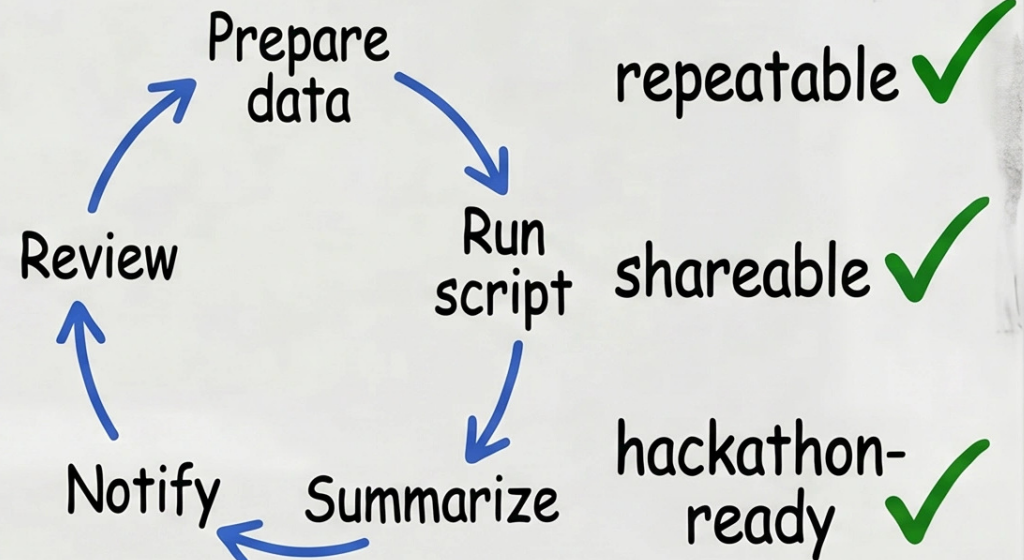


manual
rebuild 

tribal
knowledge

STANDARDIZE
THE STEPS

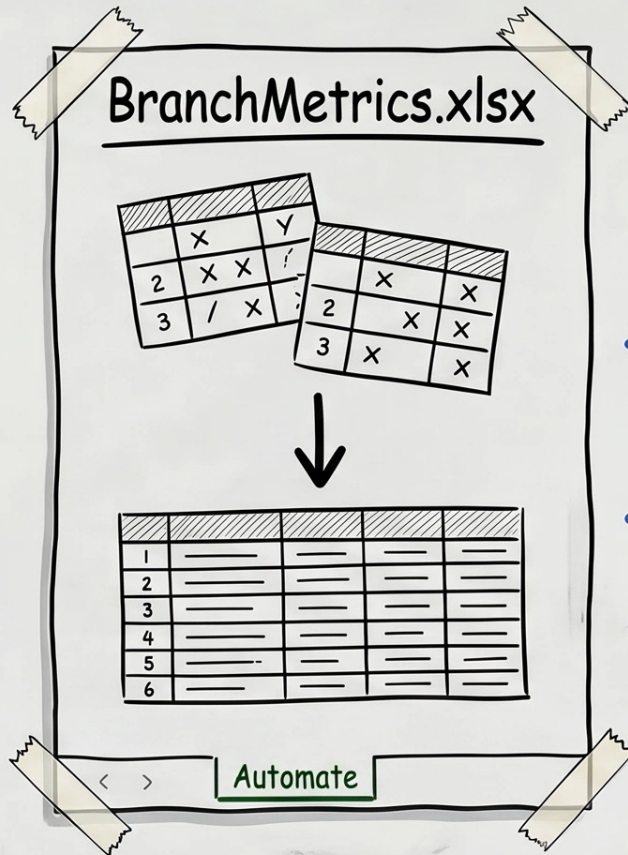
Reusable Workflow



The move: prompt → verify → script → flow → agent.

Turn the Workflow Into an Office Script

If you repeat it, record it. If you can record it, you can automate it.



Action
Recorder

Office Script

1. Format table
2. Add exception flag
3. Create summary sheet
4. Write recap block

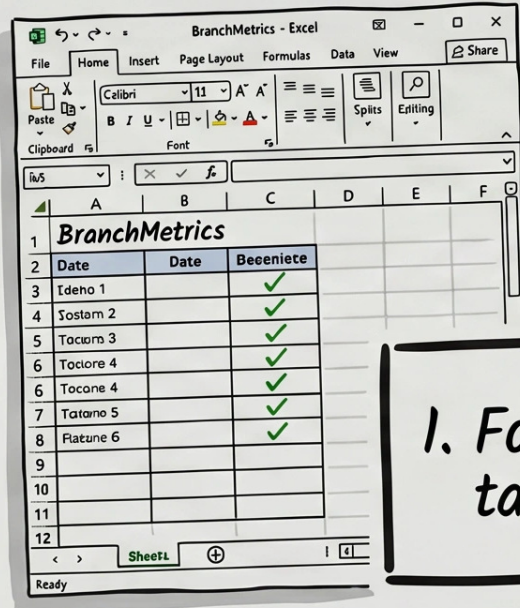
↺ Run weekly

✓ repeatable
✓ shareable
✓ auditable



Verify before
publishing

Try This: Record the Repeatable Steps



The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F
1	BranchMetrics					
2	Date	Date	Beeniete			
3	Ideno 1		✓			
4	Sostem 2		✓			
5	Tociem 3		✓			
6	Tociore 4		✓			
7	Tatame 5		✓			
8	Platune 6		✓			
9						
10						
11						
12						

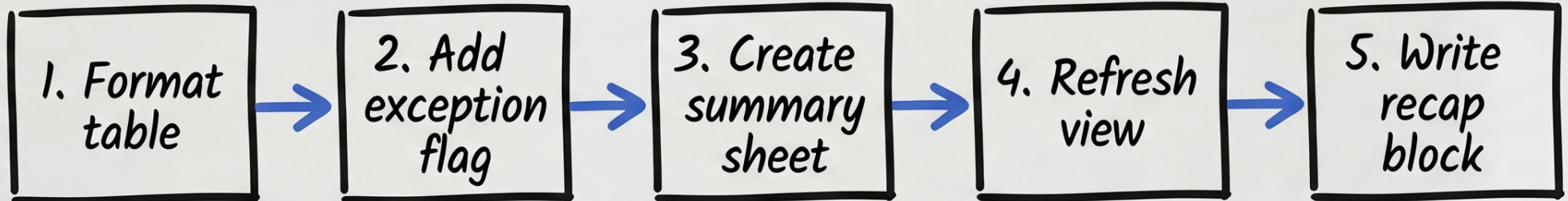


Record once

Reusable script



Run every week



- ✓ repeatable
- ✓ shareable
- ✓ hackathon-ready

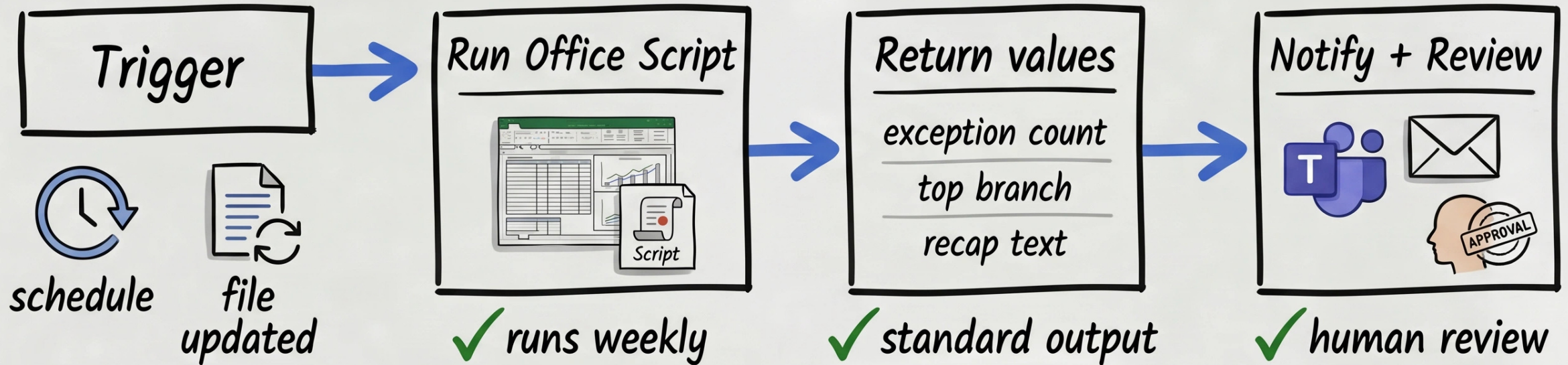


Verify before automating

3 minutes: record the steps or write the script outline.

Automate the Script With Power Automate

A script becomes powerful when a trigger can run it for you.

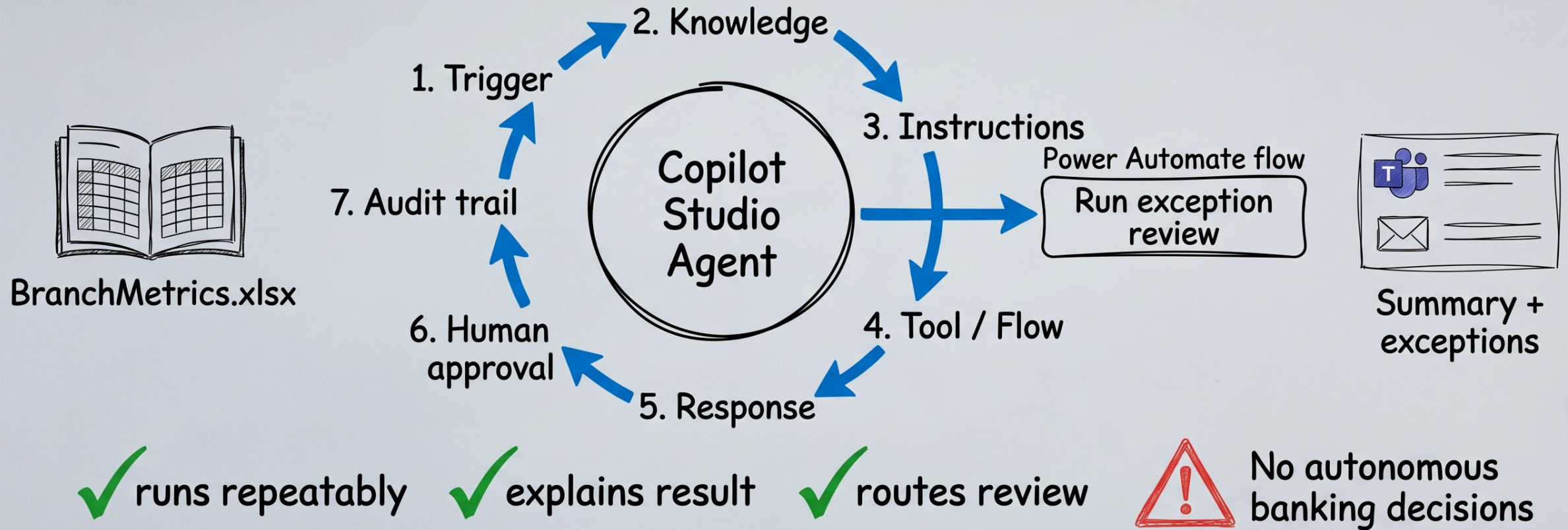


⚠ permissions + data access matter

Trigger → Script → Output → Notification → Approval 12/15

Build the Agent Around the Flow

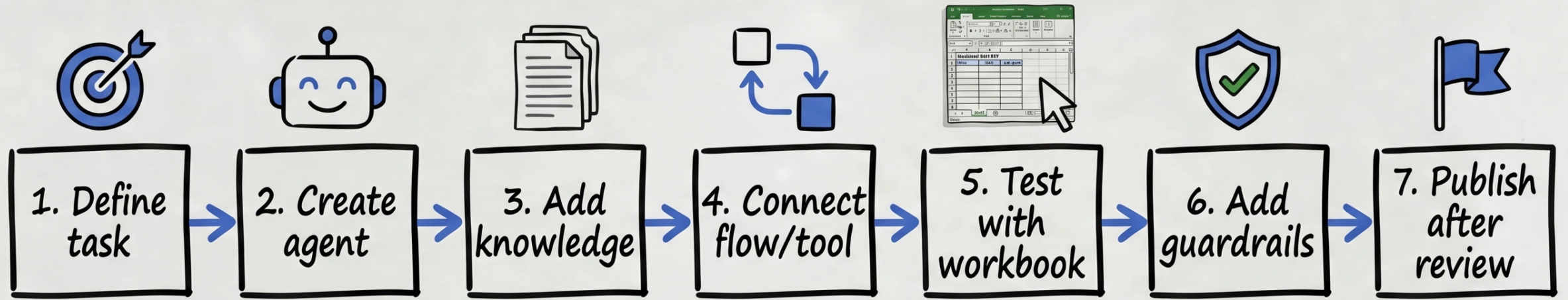
an agent does not replace the workflow; it gives people a conversational way to trigger, explain, and govern the workflow.



Agent = conversation + workflow + guardrails.

Step-by-Step Agent Build Path

Hackathon-ready sequence: turn Excel workflow into an automated agent.



✓ repeatable
✓ shareable
✓ hackathon-ready

⚠ human approval
before publish

Agent = instructions + data + tool + human review

Build the Workflow You Wish Existed

Bring one repeated spreadsheet pain point to the hackathon — then turn it into a safe, repeatable agent workflow.



The agent prepares. The human decides.

⚠ never automate judgment blindly

